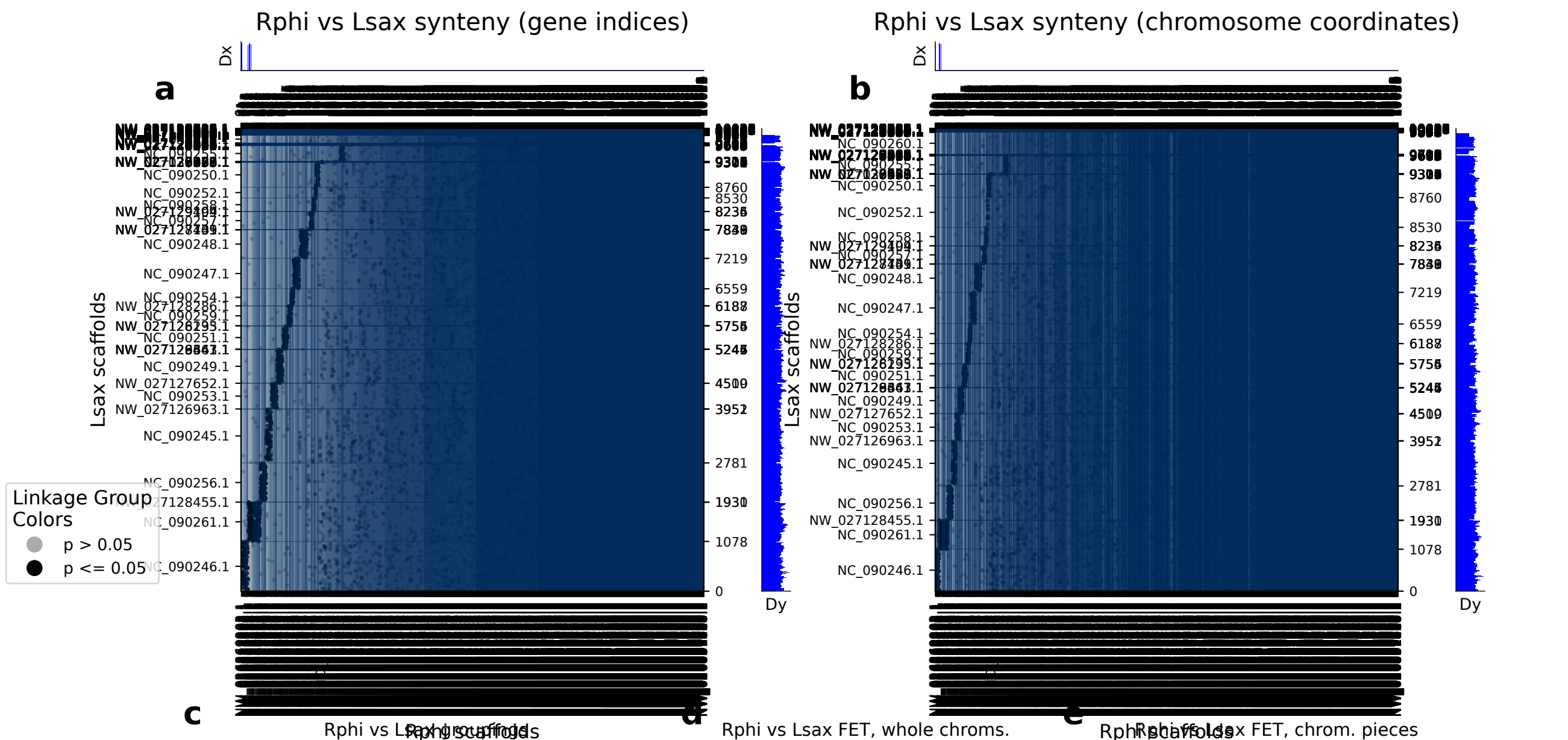


diamond: 10040 orthologs on 4112 Rphi scaffolds and 218 Lsax scaffolds.



Rphi vs L Rphi groupoids

Rphi vs Lsax FET, whole chroms.

	Rphi scaf	Lsax scaf	p (FET)	Count
0	NW_026860903.1	NC_090246.1	6.76e-14	21
1	NW_026865571.1	NC_090261.1	5.87e-13	19
2	NW_026861563.1	NC_090261.1	5.87e-13	19
3	NW_026865484.1	NC_090261.1	6.92e-13	18
4	NW_026862663.1	NC_090246.1	4.44e-10	18
5	NW_026861019.1	NC_090261.1	4.11e-11	18
6	NW_026852258.1	NC_090245.1	1.09e-10	17
7	NW_026852008.1	NC_090246.1	3.77e-09	17
8	NW_026851633.1	NC_090256.1	4.07e-10	17
9	NW_026863541.1	NC_090261.1	8.16e-10	14
10	NW_026859933.1	NC_090249.1	1.39e-09	14
11	NW_026855294.1	NC_090245.1	1.38e-04	14
12	NW_026853802.1	NC_090246.1	3.08e-07	14
13	NW_026867184.1	NC_090254.1	2.37e-12	13
14	NW_026863896.1	NC_090245.1	7.72e-06	13
15	NW_026861084.1	NC_090245.1	7.72e-06	13
16	NW_026859561.1	NC_090251.1	1.07e-11	13
17	NW_026857770.1	NC_090245.1	6.18e-07	13
18	NW_026852705.1	NC_090246.1	2.70e-06	13
19	NW_026852000.1	NC_090259.1	1.70e-11	13
20	NW_026862783.1	NC_090246.1	1.49e-04	12
21	NW_026858503.1	NC_090248.1	2.44e-09	12
22	NW_026854960.1	NC_090249.1	1.92e-08	12
23	NW_026865967.1	NC_090249.1	1.81e-05	11
24	NW_026865813.1	NC_090245.1	4.63e-05	11
25	NW_026859998.1	NC_090249.1	2.99e-06	11
26	NW_026859790.1	NC_090261.1	1.54e-05	11
27	NW_026859620.1	NC_090261.1	1.54e-05	11
28	NW_026859418.1	NC_090245.1	4.97e-04	11
29	NW_026855828.1	NC_090249.1	2.99e-06	11
30	NW_026853170.1	NC_090246.1	1.20e-03	11
31	NW_026852998.1	NC_090247.1	5.67e-06	11
32	NW_026866451.1	NC_090249.1	3.80e-05	10
33	NW_026866098.1	NC_090246.1	1.77e-04	10
34	NW_026864957.1	NC_090261.1	1.65e-05	10
35	NW_026864352.1	NC_090249.1	3.80e-05	10
36	NW_026862956.1	NC_090261.1	9.31e-04	10
37	NW_026862474.1	NC_090247.1	3.02e-04	10
38	NW_026861872.1	NC_090250.1	1.71e-07	10
39	NW_026859313.1	NC_090247.1	1.27e-06	10
40	NW_026858000.1	NC_090247.1	1.27e-06	10
41	NW_026856137.1	NC_090261.1	1.68e-04	10
42	NW_026852468.1	NC_090261.1	1.65e-05	10
43	NW_026867313.1	NC_090248.1	1.09e-05	9
44	NW_026866838.1	NC_090256.1	1.78e-03	9
45	NW_026866837.1	NC_090261.1	1.82e-03	9
46	NW_026865475.1	NC_090251.1	8.98e-05	9
47	NW_026864022.1	NC_090251.1	1.79e-06	9
48	NW_026863691.1	NC_090248.1	1.03e-04	9
49	NW_026863645.1	NC_090257.1	9.07e-06	9
50	NW_026861840.1	NC_090251.1	2.85e-03	9
51	NW_026861574.1	NC_090247.1	1.84e-04	9
52	NW_026861405.1	NC_090253.1	3.99e-05	9
53	NW_026860451.1	NC_090249.1	2.46e-03	9
54	NW_026859959.1	NC_090256.1	1.93e-04	9
55	NW_026859757.1	NC_090249.1	4.78e-04	9
56	NW_026859085.1	NC_090246.1	1.66e-03	9
57	NW_026857866.1	NC_090256.1	1.78e-03	9
58	NW_026857461.1	NC_090245.1	3.46e-03	9
59	NW_026857221.1	NC_090253.1	2.09e-04	9
60	NW_026857083.1	NC_090250.1	3.07e-05	9
61	NW_026855197.1	NC_090253.1	3.38e-02	9
62	NW_026854764.1	NC_090247.1	1.95e-05	9
63	NW_026854098.1	NC_090248.1	1.09e-05	9

RphRph¹⁶Lox FET, chrom. pieces

	Rphi scaf	Lsax scaf	p (FET)	Count
0	NW_026860903.1 : 0-end	NC_090246.1 : 0-end	6.76e-1421	
1	NW_026865571.1 : 0-end	NC_090261.1 : 0-end	5.87e-1319	
2	NW_026861563.1 : 0-end	NC_090261.1 : 0-end	5.87e-1319	
3	NW_026865484.1 : 0-end	NC_090261.1 : 0-end	6.92e-1318	
4	NW_026862663.1 : 0-end	NC_090246.1 : 0-end	4.44e-1018	
5	NW_026861019.1 : 0-end	NC_090261.1 : 0-end	4.11e-1118	
6	NW_026852258.1 : 0-end	NC_090245.1 : 0-end	1.09e-1017	
7	NW_026852008.1 : 0-end	NC_090246.1 : 0-end	3.77e-0917	
8	NW_026851633.1 : 0-end	NC_090256.1 : 0-end	4.07e-1017	
9	NW_026863541.1 : 0-end	NC_090261.1 : 0-end	8.16e-1014	
10	NW_026859933.1 : 0-end	NC_090249.1 : 0-end	1.39e-0914	
11	NW_026855294.1 : 0-end	NC_090245.1 : 0-end	1.38e-0414	
12	NW_026853802.1 : 0-end	NC_090246.1 : 0-end	3.08e-0714	
13	NW_026867184.1 : 0-end	NC_090254.1 : 0-end	2.37e-1213	
14	NW_026863896.1 : 0-end	NC_090245.1 : 0-end	7.72e-0613	
15	NW_026861084.1 : 0-end	NC_090245.1 : 0-end	7.72e-0613	
16	NW_026859561.1 : 0-end	NC_090251.1 : 0-end	1.07e-1113	
17	NW_026857770.1 : 0-end	NC_090245.1 : 0-end	6.18e-0713	
18	NW_026852705.1 : 0-end	NC_090246.1 : 0-end	2.70e-0613	
19	NW_026852000.1 : 0-end	NC_090259.1 : 0-end	1.70e-1113	
20	NW_026862783.1 : 0-end	NC_090246.1 : 0-end	1.49e-0412	
21	NW_026858503.1 : 0-end	NC_090248.1 : 0-end	2.44e-0912	
22	NW_026854960.1 : 0-end	NC_090249.1 : 0-end	1.92e-0812	
23	NW_026859367.1 : 0-end	NC_090249.1 : 0-end	1.81e-0511	
24	NW_026865813.1 : 0-end	NC_090245.1 : 0-end	4.63e-0511	
25	NW_026859998.1 : 0-end	NC_090249.1 : 0-end	2.99e-0611	
26	NW_026859790.1 : 0-end	NC_090261.1 : 0-end	1.54e-0511	
27	NW_026859620.1 : 0-end	NC_090261.1 : 0-end	1.54e-0511	
28	NW_026859418.1 : 0-end	NC_090245.1 : 0-end	4.97e-0411	
29	NW_026855828.1 : 0-end	NC_090249.1 : 0-end	2.99e-0611	
30	NW_026853170.1 : 0-end	NC_090246.1 : 0-end	1.20e-0311	
31	NW_026852988.1 : 0-end	NC_090247.1 : 0-end	5.67e-0611	
32	NW_026866451.1 : 0-end	NC_090249.1 : 0-end	3.80e-0510	
33	NW_02686098.1 : 0-end	NC_090246.1 : 0-end	1.77e-0410	
34	NW_026864957.1 : 0-end	NC_090261.1 : 0-end	1.65e-0510	
35	NW_026864352.1 : 0-end	NC_090249.1 : 0-end	3.80e-0510	
36	NW_026862956.1 : 0-end	NC_090261.1 : 0-end	9.31e-0410	
37	NW_026862474.1 : 0-end	NC_090247.1 : 0-end	3.02e-0410	
38	NW_026861872.1 : 0-end	NC_090250.1 : 0-end	1.71e-0710	
39	NW_026859313.1 : 0-end	NC_090247.1 : 0-end	1.27e-0610	
40	NW_026858000.1 : 0-end	NC_090247.1 : 0-end	1.27e-0610	
41	NW_026856137.1 : 0-end	NC_090261.1 : 0-end	1.68e-0410	
42	NW_026852468.1 : 0-end	NC_090261.1 : 0-end	1.65e-0510	
43	NW_026867313.1 : 0-end	NC_090248.1 : 0-end	1.09e-059	
44	NW_026866838.1 : 0-end	NC_090256.1 : 0-end	1.78e-039	
45	NW_026866837.1 : 0-end	NC_090261.1 : 0-end	1.82e-039	
46	NW_026865475.1 : 0-end	NC_090251.1 : 0-end	8.98e-059	
47	NW_026864022.1 : 0-end	NC_090251.1 : 0-end	1.79e-069	
48	NW_026863691.1 : 0-end	NC_090248.1 : 0-end	1.03e-049	
49	NW_026863645.1 : 0-end	NC_090257.1 : 0-end	9.07e-069	
50	NW_026861840.1 : 0-end	NC_090251.1 : 0-end	2.85e-039	
51	NW_026861574.1 : 0-end	NC_090247.1 : 0-end	1.84e-049	
52	NW_026861405.1 : 0-end	NC_090253.1 : 0-end	3.99e-059	
53	NW_026860451.1 : 0-end	NC_090249.1 : 0-end	2.46e-039	
54	NW_026859959.1 : 0-end	NC_090256.1 : 0-end	1.93e-049	
55	NW_026859757.1 : 0-end	NC_090249.1 : 0-end	4.78e-049	
56	NW_026859085.1 : 0-end	NC_090246.1 : 0-end	1.66e-039	
57	NW_026857866.1 : 0-end	NC_090256.1 : 0-end	1.78e-039	
58	NW_026857461.1 : 0-end	NC_090245.1 : 0-end	3.46e-039	
59	NW_026857221.1 : 0-end	NC_090253.1 : 0-end	2.09e-049	
60	NW_026857083.1 : 0-end	NC_090250.1 : 0-end	3.07e-059	
61	NW_026855197.1 : 0-end	NC_090253.1 : 0-end	3.38e-029	
62	NW_026854764.1 : 0-end	NC_090247.1 : 0-end	1.95e-059	
63	NW_026854098.1 : 0-end	NC_090248.1 : 0-end	1.09e-059	

(a) depicts the chromosomes/scaffolds of Rphi (x-axis) plotted against the chromosomes/scaffolds of Lsax (y axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10040 orthologs found between 4112 Rphi scaffolds and 218 Lsax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved syntenies resolves early events⁵In vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830.

(b) depicts the same information as panel a, but it is plotted in the organisms' chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (c) shows chromosome-scale significance values. This table shows the same information as c, but does not show genomic information. It shows the FET p-values for single arms of chromosomes correlated with other regions.

	Rphi Chromosome Index	Lsax Chromosome Index	FET P-value	Dx	Dy	Chromosome Scale Significance
73	NW_026865464.1	NC_090256.1	1.91e-05.8	-	-	1.91e-05.8
74	NW_026865227.1	NC_090257.1	0.0000000099.83	-	-	0.0000000099.83
75	NW_026865060.1	NC_090245.1	2.98e-02.8	-	-	2.98e-02.8
76	NW_026865440.1	NC_090247.1	2.98e-02.8	-	-	2.98e-02.8